# THARUN G

**Cybersecurity | SOC & Application Security | CTF Player**

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## SUMMARY

Third-year CSE student focused on offensive and defensive security. Built an autonomous SIEM/SOAR pipeline and an end-to-end encrypted messenger, plays and authors web-exploitation CTFs, and practises regularly on TryHackMe. Seeking a security internship or entry-level role (SOC, VAPT, application security).

## EDUCATION

**B.E. Computer Science and Engineering** | National Engineering College, Kovilpatti 2024 – 2028 (Expected)

CGPA: 6.8 / 10

## SKILLS

**Security:** Web Application Security, Network Security, Vulnerability Assessment, Threat Modelling, Ethical Hacking, Burp Suite, Nmap, MITRE ATT&CK

**Detection & Response:** SIEM/SOAR, OpenSearch, Sigma rules, OCSF, TheHive, Shuffle

**Programming & Tools:** Python, TypeScript, Node.js, SQL, Docker, Linux (Kali/Ubuntu), Git

## PROJECTS

**SentinelAI: Autonomous SIEM & SOAR Assistant** | Python, LangGraph, Ollama, Redpanda, OpenSearch, Sigma, TheHive, Shuffle, Docker

- Built a containerized SIEM/SOAR pipeline that normalizes endpoint, network and cloud logs to the OCSF schema with Vector and Redpanda (Kafka) and indexes them in OpenSearch.
- Compiled Sigma rules into OpenSearch queries to detect MITRE ATT&CK techniques such as brute force and PowerShell download activity.
- Built a LangGraph triage agent (local Llama 3.1 via Ollama) with deterministic fallbacks that scores alerts and triggers allow-listed playbooks (host isolation, IP block, user disable) through TheHive 4 and Shuffle.

**Orion AI: Local-First Personal AI Assistant** | Python, Rust (PyO3), FastAPI, Docker, Wasm github.com/AstraDev-Labs/Orion-AI

- Built an offline autonomous assistant with guardrails for agent safety: PII masking, SSRF network filters, prompt-injection detection and persistent audit logging.
- Sandboxed agent code execution with Docker/Wasm and required human approval for sensitive system operations; implemented the security scanning modules in Rust via PyO3.

**Prism: End-to-End Encrypted Messaging** | Next.js, Socket.IO, MongoDB, Web Crypto API

- Designed a messenger in which the server cannot decrypt messages, using client-side RSA/AES encryption with the Web Crypto API.
- Combined the encryption with Socket.IO real-time delivery while the server stores only encrypted payloads.

**Aegis Toolkit: Security Assessment Suite (in progress)** | Python, Tkinter, Nmap

- Combining Nmap-based network scanning, hash calculation and encoding/decoding tools in one desktop interface to speed up routine assessments.

## CTF & HANDS-ON PRACTICE

**Handle:** ragefrost. Focus areas: web exploitation, SSRF chaining, JWT and prototype-pollution attacks, authentication bypass, Active Directory attacks.

**Events:** GenCysCTF 2.0, CloudSEK CTFx (Black Vault CTF), MythX National CTF (KIET).

**Challenge author:** Built "Merge Conflict" for crimsonCTF: a Node.js/Express chain of .git leak, JWT algorithm confusion, prototype pollution and EJS RCE.

**TryHackMe:** hands-on practice on guided paths and labs, tryhackme.com/p/Ragefrost

## EXPERIENCE

**Web Development Intern** | Prodigy InfoTech Aug 2025

- Developed frontend and backend features for client web projects; received a letter of recommendation for problem-solving and reliability.

## CERTIFICATIONS

- Cybersecurity Bootcamp: Software Security and Vulnerability Assessment, IIITDM Kancheepuram
- API Security Training Certificate, APISEC CON (Feb 2026)
- Discovering Cybersecurity (Udacity & Accenture, 2026)